Solar PV sizing — panels and inverters

REV —

Villa Complex 415V LV Network

Array

Sized by	Target 50 kWp
Panel	Mono PERC 550 W (generic) — 550 W, Voc 49.6 V, Vmp 41.7 V, Isc 14 A, 2.278 × 1.134 m
Array	**91 panels = 50.05 kWp** in 7 strings of 13
Panel area / roof needed	235 m² / ≈ 392 m²

Inverters

Inverter	String inverter (generic) — 50 kW, 3-phase, max DC 1100 V, MPPT 200–1000 V, 4 MPPTs × 40 A
Selected	**1 × 50 kW = 50 kW AC**, DC/AC 1.00
AC connection	72 A → 100 A breaker (3-phase, 415 V line-to-line)
SLD connection	On MDB-1: 100 A breaker, 4C × 25 mm², 30 m (SLD sizing: I ÷ 0.85, cable for Iz and voltage drop)

String design (IEC 62548)

Site temperatures	min 10 °C, max 48 °C → max cell 79 °C (NOCT 45 °C)
Temperature coefficients	Voc -0.27 %/°C (maximum DC voltage) · Vmp -0.4 %/°C (MPPT window) — **ESTIMATED: no datasheet Vmp coefficient; typical crystalline-silicon value, MPPT checks not verified** · Pmax -0.35 %/°C (yield). Linear approximation $V(T) = V_{STC} \times (1 + \text{coefficient} \times (T - 25\ °C))$.
String Voc at 10 °C	13 × 51.61 = 671 V ≤ 1100 V **OK**
String Vmp at 79 °C	13 × 32.65 = 424 V ≥ 200 V OK (estimated)
String Vmp at 10 °C	13 × 44.2 = 575 V ≤ 1000 V OK (estimated)
MPPT current	2 × 1.25 × 14 A = 35 A ≤ 40 A **OK**
Panels per string that fit	7 – 21

Energy

Peak sun hours	5.8 kWh/m²/day
Losses	temperature 12.2 % · soiling 5 % · mismatch 2 % · DC cable 1.5 % · AC cable 1 % · inverter 1.8 %
Performance ratio	78.2 %
Yield	**227 kWh/day · 82,890 kWh/year** (1,656 kWh/kWp)
Savings	31,498 per year at 0.38 per kWh
CO_2 avoided	33.2 t/year at 0.4 kg/kWh

MPPT window not verified: no Vmp temperature coefficient for the panel — estimated at -0.4 %/°C (typical crystalline silicon). Enter the datasheet value.

Yield = kWp × peak sun hours × performance ratio (first estimate; confirm with a simulation such as PVsyst for the final design). Panel and inverter values from the datasheets entered.